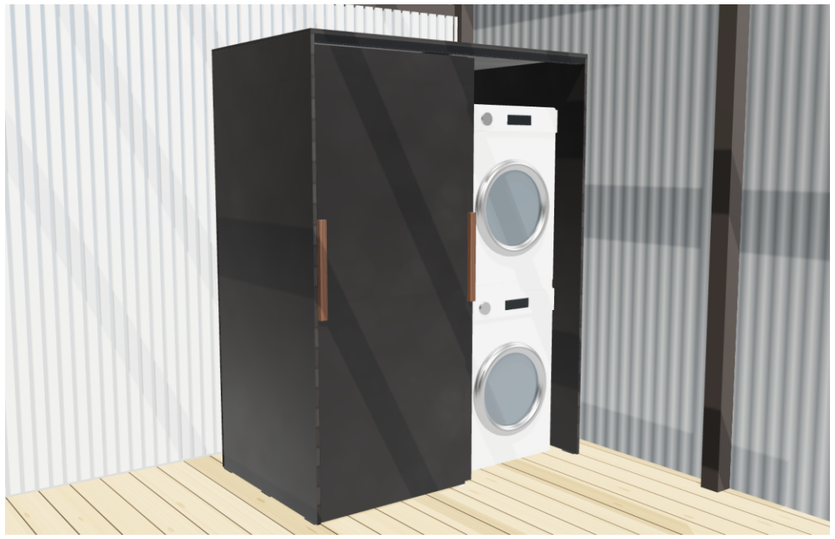
Outdoor appliance cabinet: fridge + stacked washer/dryer





A plain black formply box: flush top, the fridge door flush with the sides, Cowdroy Quick-Fix door gear from Bunnings, no front vents, and two short American walnut pulls.

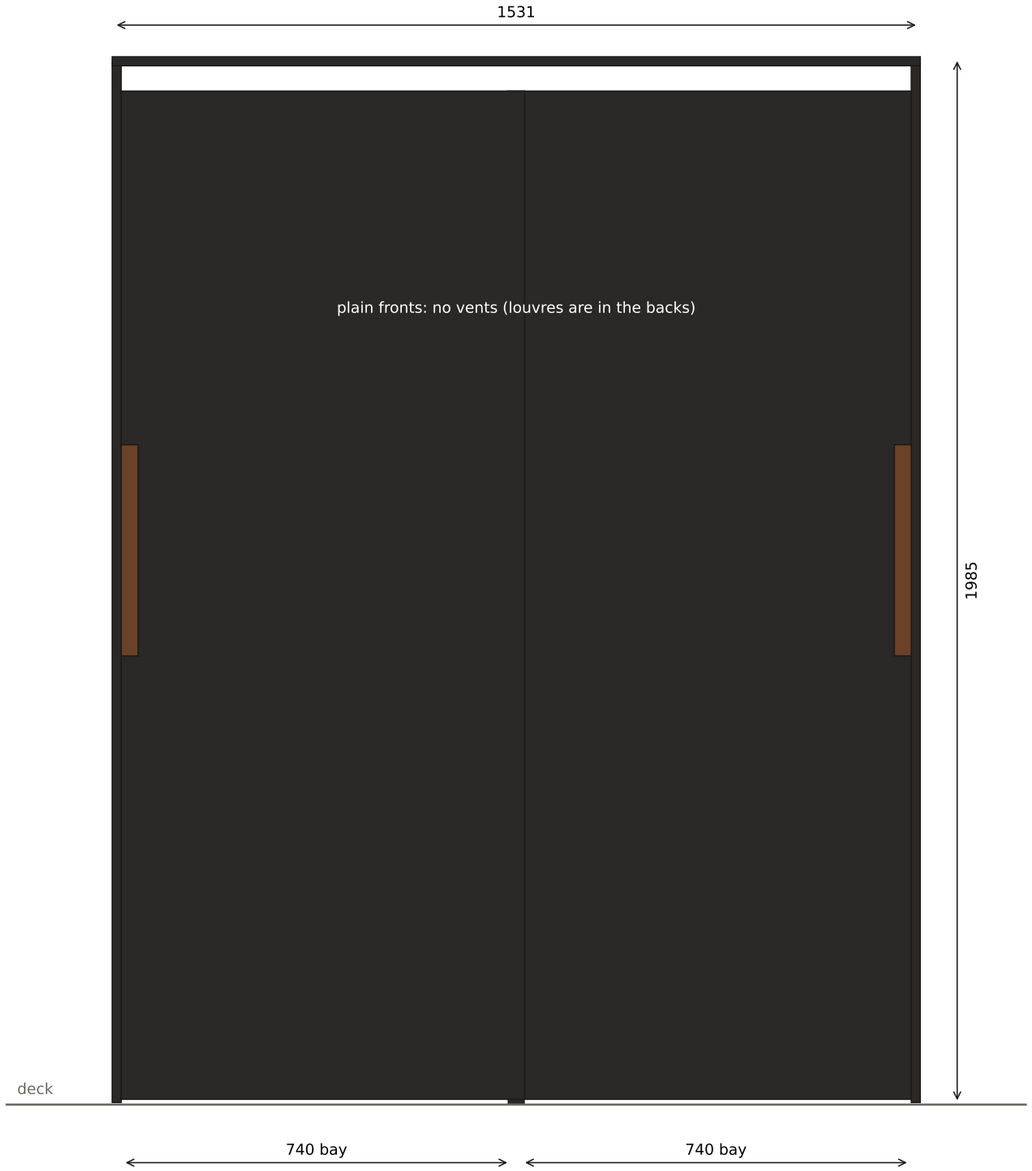
1531 W × 900 D × 1985 H. Two 740 clear bays. 8 CNC parts on 8 sheets of 2400 × 1200 × 17.

The fridge breathes through flush louvred openings low and high in the hidden back panels (about 200 cm2 clear each level). Run the dryer with the laundry door slid open. The box sits on 4 mm ASA pads, held down by two stainless angles into the deck joists.

22 of 22 design checks pass (sheet 6). Open items must be closed before the cut.

General arrangement

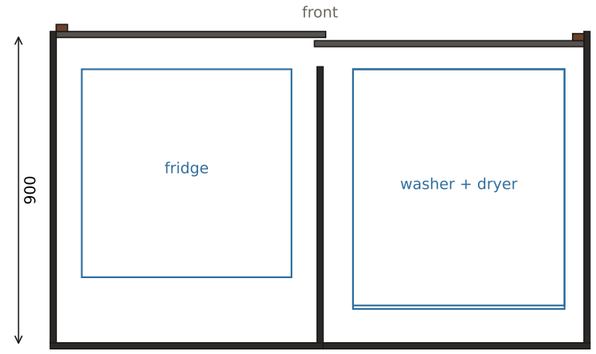
Front elevation (doors closed)



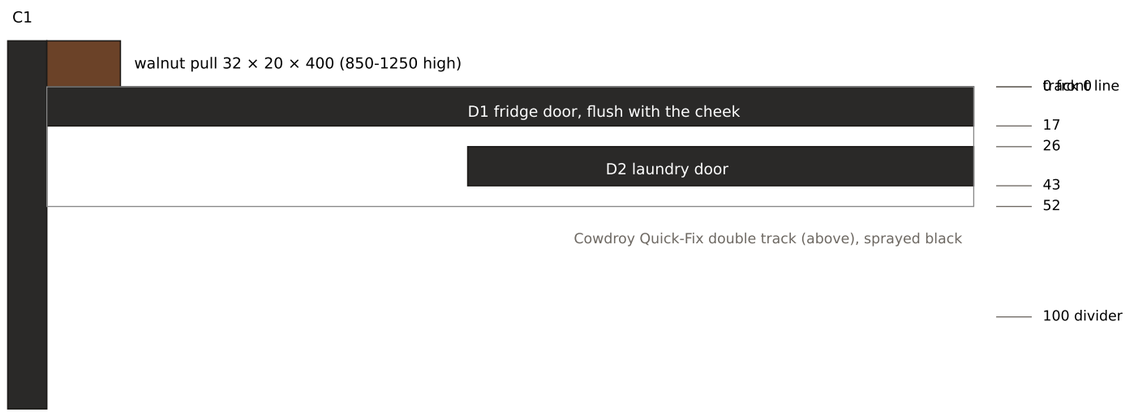
Side elevation (studio cheek, outside)



Plan (top removed)

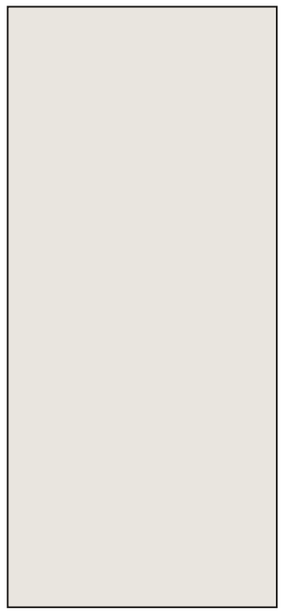


Section through the doors and tracks (plan, enlarged)



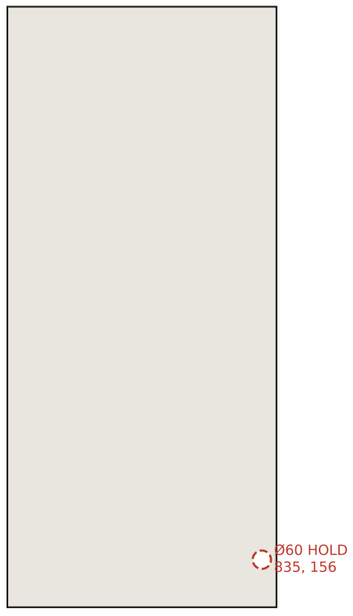
CNC part drawings

C1 Left cheek 883 × 1970 ×1



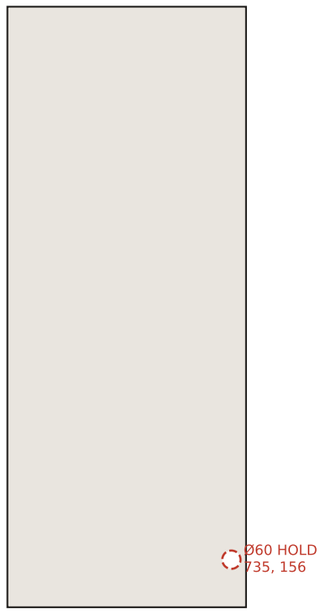
X = 0 at the cabinet front edge, Y = 0 at the bottom edge.

C3 Studio cheek 883 × 1970 ×1



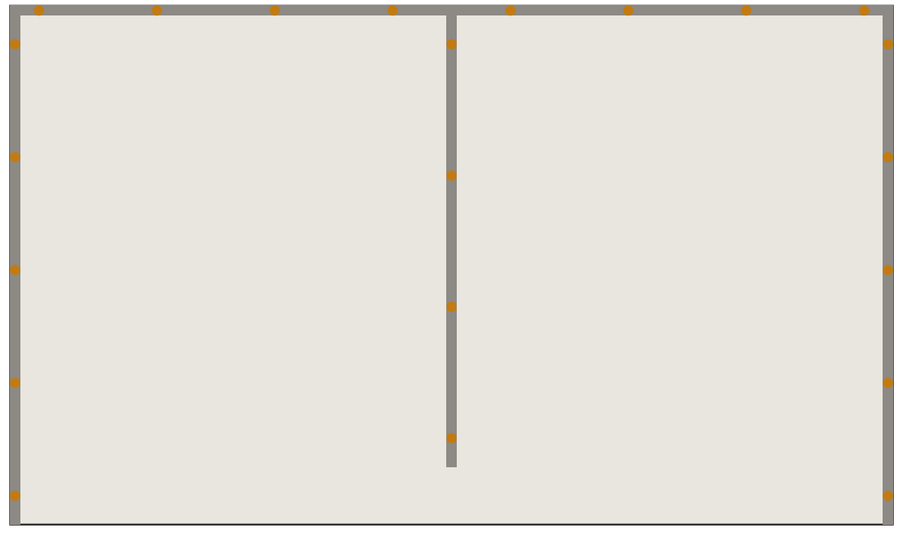
X = 0 at the cabinet front edge (mirror view: routed from the outside face).

C2 Divider 783 × 1970 ×1



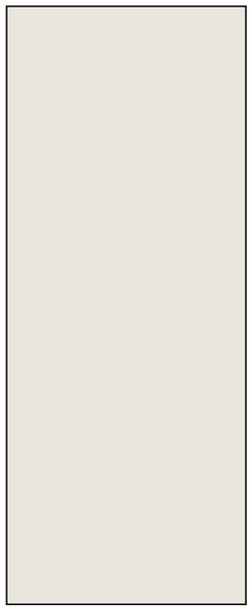
X = 0 at the front edge.

C4 Top 1531 × 900 ×1



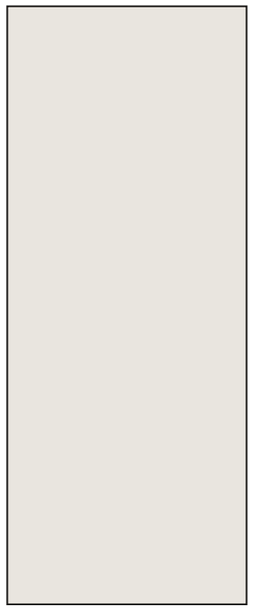
Routed on the UNDERSIDE: 6 mm housings for every vertical. Drilled through 3.5 for the screws (DRILL layer). X = 0 at the kitchen-side edge, Y = 0 at the front edge.

D1 Fridge door 764.5 × 1910 ×1



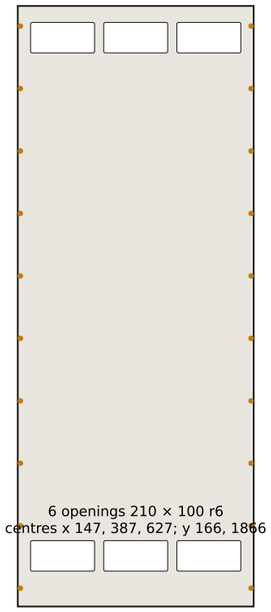
Plain panel. Two ASA top shoes on the top edge carry the Quick-Fix wheel brackets.

D2 Laundry door 764.5 × 1910 ×1



Plain panel. Two ASA top shoes on the top edge carry the Quick-Fix wheel brackets.

C5 Back, fridge bay 774 × 1970 ×1



6 louvre openings 210 x 100 r6 and 3.5 screw holes. Drawn from the INSIDE face (as seen from the front of the cabinet), X = 0 at the kitchen-side edge.

C6 Back, laundry bay: as C5.
C4 is routed on its UNDERSIDE: 6 deep housings (grey) for C1, C2, C3, the backs and the head rail.
Cheek housings are 0.6 over the panel for glue and fit.

Cut list and indicative nests

Code	Part	Size	Qty	Notes
C1	Left cheek	883 × 1970 × 17	1	X = 0 at the cabinet front edge, Y = 0 at the bottom edge.
C2	Divider	783 × 1970 × 17	1	X = 0 at the front edge.
C3	Studio cheek	883 × 1970 × 17	1	X = 0 at the cabinet front edge (mirror view: routed from the outside face).
C4	Top	1531 × 900 × 17	1	Routed on the UNDERSIDE: 6 mm housings for every vertical. Drilled through 3.5 for the screws (DRILL […]
C5	Back, fridge bay	774 × 1970 × 17	1	6 louvre openings 210 x 100 r6 and 3.5 screw holes. Drawn from the INSIDE face (as seen from the front […]
C6	Back, laundry bay	757 × 1970 × 17	1	As C5, drawn from the INSIDE face, X = 0 at the edge that butts C5. Not symmetric: screw holes are on […]
D1	Fridge door	764.5 × 1910 × 17	1	Plain panel. Two ASA top shoes on the top edge carry the Quick-Fix wheel brackets.
D2	Laundry door	764.5 × 1910 × 17	1	Plain panel. Two ASA top shoes on the top edge carry the Quick-Fix wheel brackets.

S1 2400 × 1200 × 17



S2 2400 × 1200 × 17



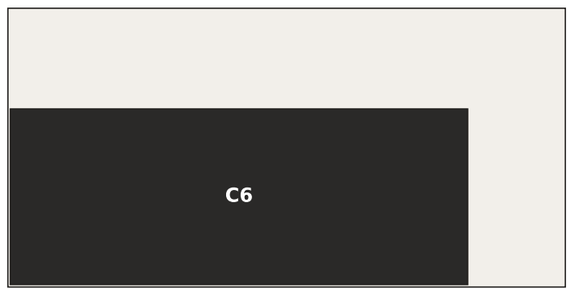
S3 2400 × 1200 × 17



S4 2400 × 1200 × 17



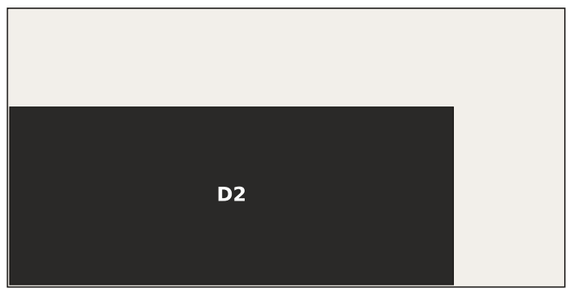
S5 2400 × 1200 × 17



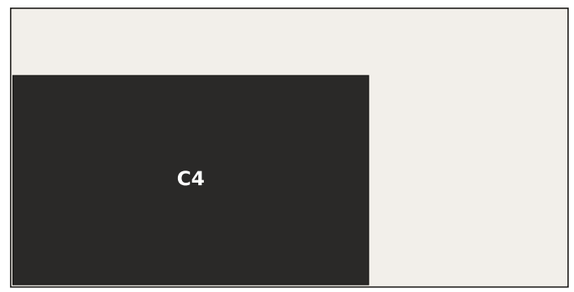
S6 2400 × 1200 × 17



S7 2400 × 1200 × 17



S8 2400 × 1200 × 17



Build steps and shopping list

1. Confirm before cutting

Confirm the washer and dryer models and their stacking kit (heat-pump or condenser dryer). Buy the Cowdroy Quick-Fix set from Bunnings first and check its track height (30) and the gap between the two door lines (about 26) against the plans. Measure the power point position along the studio wall and the gaps to the slat screen and studio wall.

2. Pick-up check

Eight parts, all 17 mm. Each is labelled with its code and an arrow to its top and front on the inside face. Check the 3.5 mm screw holes are drilled through the top (C4) and both backs (C5, C6), the six louvre openings in each back and the two lead holes. Keep the sheets flat and dry until assembly.

3. Seal, spray and print

Two coats of exterior edge sealer on every cut edge, door bottoms and tops most of all. Etch-prime and spray the Quick-Fix track satin black. Print the ASA parts. Cut a 740 spacer from an offcut for setting the bays.

4. Pads on the deck

Mark the footprint. Bed the thirteen 4 mm ASA edge pads in sealant, one 10g x 25 each: three under each cheek, three under the divider, four under the backs.

5. Track, then the verticals, upside down

Lay the top film-face down on blankets on a flat floor or trestles, housings up. Screw the black track into the underside first, front face in line with the front edge: it is easy now and overhead later. Run PU sealant in the cheek and divider housings and stand C1, C2 and C3 in them, inside faces in. Set both bays with the 740 spacer and tack a batten across the free ends to hold them.

6. Backs: this squares the box

Run sealant along the rear edges and the back rebate. Stand C5 in the rebate, flush with the kitchen cheek's outside face, then C6 butted against it. Through each CNC hole in the backs, drill a 3 mm pilot about 30 deep into the edge behind, then drive a 10g x 45 stainless decking screw until the head sits just flush. The heads end up in neat rows on the back, facing the slat screen. Fit the black joint angle inside the laundry bay (C6 to the divider), then push a louvre insert into each opening and staple mesh across the inside.

7. Top screws

Two people: tip the box onto its back on blankets so the top is at bench height. Pilot through each hole in the top into the edge below (cheeks, divider, backs) and screw as before. All of these heads sit on top of the cabinet, 1985 up, where nobody sees them.

8. Stand it up and hold it down

Two people: stand the box upright and walk it onto the pads. Check plumb and the diagonals. Screw the two stainless hold-down angles inside the cheeks into the deck joists and the cheeks. Remove the batten.

9. Doors and pulls

Screw the walnut pulls on from behind (left edge of D1, right edge of D2, 850-1250 up). Glue and screw two ASA top shoes onto the top edge of each door (leg down the back face, 110 in from each end), then screw a Quick-Fix wheel bracket down through each shoe into the door edge, hang D2 on the rear line first, then D1 on the front, and adjust to 10 off the deck with D1 flush with the cheeks. Fit a stop so D1 parks 40 short of the studio cheek.

10. Floor guide

Screw the Quick-Fix nylon floor guide to the deck where the doors cross (about 765 from the kitchen side). Run both doors end to end.

11. Grommets and leads

Fit the split grommets in the two low lead holes (160 above the deck) and run each original lead separately to the power point. No powerboard, nothing in a door track or under a machine foot.

12. Appliances in

Roll the fridge in, door hinge on the left, 107 back from the front. Stand the washer, fit the maker's stacking kit and lift the dryer on. Level all three on their own feet; the cabinet carries none of them.

Type	Item	Qty	Use
Sheet	Black film-faced formply 2400 x 1200 x 17, exterior (A-bond)	8 sheets	CNC supplier supplies and cuts (Cut list tab)
Sheet	Exterior edge sealer, black	1 L	Two coats on every cut edge and the door grooves
Sheet	PU sealant/adhesive, black (Sikaflex-11FC or similar)	4 x 300 ml	Housings, angles, louvres, pads
Hardware	Cowdroy 2400mm Quick-Fix Door Track Set (Bunnings I/N 3981776): double J track, wheel sets, nylon floor guide	1 set, track cut to 1	Top-hung bi-passing, 25 kg per door. Rated for 19 mm+ doors: the ASA top shoes adapt our 17 mm doors. Uses 4 of the 6
Hardware	Etch primer + satin black spray paint	1 each	Track
Hardware	Aluminium angle 25 x 25 x 3, black anodised	1 x 1520	Inside the laundry bay, C6 to the divider
Hardware	Stainless angle 50 x 40 x 3	2 x 300	Hold-downs inside the cheeks
Hardware	Stainless insect mesh	1 m x 300	Inside the twelve back openings
Timber	American walnut 32 x 20 x 400, from workshop offcuts	2	Door pulls
Timber	Exterior hardwood oil (e.g. Cutek or Intergrain UltraDeck Oil)	250 ml	Walnut pulls; seal end grain
Fixings	Stainless decking screws, 10g x 45, T25 Torx	1 box of 100	All carcass joints: 52 screws through CNC-drilled holes in the top and backs
Tools	3 mm drill bit (long) + T25 driver bit	1 each	Pilot through each CNC hole into the edge behind
Fixings	8g x 20 stainless countersunk	30	Joint angle and hold-down angles
Fixings	10g x 50 stainless countersunk	4	Hold-downs into the deck joists
Fixings	10g x 25 stainless countersunk	16	Pads
Fixings	6g x 16 stainless pan	24	Louvre inserts
Fixings	8g x 30 stainless countersunk	6	Pulls, from behind
Fixings	Stainless staples, 10 mm	1 box	Mesh
Print	ASA filament, black	about 2.6 kg	Prints tab: 4 STLs, 31 prints

Checks, open items and ASA prints

	Check	Result
Pass	Fridge side clearance (maker 50)	72.5 each side
Pass	Fridge back clearance (maker 50)	186 to the back panel: the chimney behind the fridge
Pass	Fridge top clearance (maker 100)	183 to the top; the tracks sit in front of the fridge face
Pass	Stack under the top	238 above the dryer top
Pass	Laundry passes the open door	opening 841.5-1514, machines 859.0-1459.0
Pass	Rear pull clear of the closed front door	28 mm finger gap
Pass	Fridge door parks clear of the laundry pull	parked fridge door ends 8 short of the pull
Pass	Fridge opening with the door parked short	opening 17-709.5, fridge 89.5-684.5
Pass	Rear pull in the gap between the doors	pull face 6 behind the front line
Pass	Quick-Fix limits (25 kg per door; 19 mm min met by th	764.5 x 1910 x 17, about 16 kg
Pass	Front door shoe leg clears the rear door	5 mm between the leg and the rear door
Pass	Track fits the opening	1497 opening; Quick-Fix suits openings up to 1800
Pass	Rear track clear of the divider	48 mm
Pass	No screw heads on show	52 screws: all through the top (1985 up, above eye level) or the backs (facin
Pass	Screws bite into edges	45 screw through 17 leaves 28 in the edge
Pass	Screws land on edge centrelines	every hole is 8.5 from the panel face it enters, so 6.7 of ply each side of a 10
Pass	Leads exit near the power point	lead holes centred 160 above the deck; power point about 100
Pass	Fridge vents (Liebherr/Electrolux built-in rule: 200 cm2	per level per bay: 3 openings, 630 cm2 gross, about 207 net through louvre
Pass	No straight path in for rain	louvre blades 45 degrees down and out, 27 deep at 13 pitch: each blade ove
Pass	High vents above the appliances	high openings start 1820; fridge 1785, stack 1730
Pass	Doors clear of the deck	10 above the deck
Pass	Fridge door opens 115 deg past the cheek	closest 24 mm at 115 deg (simplified slab, left hinge)

Open item	Before the cut
Washer and dryer models	Provisional 600 x 680 x 850 and 600 x 670 x 850. Confirm both models and the maker's stacking kit; the dryer must be heat-pump or condenser.
Dryer air	Run the dryer with the laundry door slid open. Makers say a cupboard dryer needs louvred door openings or the door open while it runs (LG asks for two door openings of 387 cm2). The back vents help but are not a substitute.
Quick-Fix bracket on the sh	Cowdroy rates Quick-Fix for 19 mm+ doors; the 21 mm ASA shoe gives the bracket a seat that wide on our 17 mm doors. Buy the set first: check the bracket's footprint fits the 80 x 21 seat, the track height (30) and the gap between door lines (about 26) before the doors are released.
Hardware outdoors	Quick-Fix is sold for indoor wardrobes. Under cover it should be fine; spray the wheels with dry PTFE lube once a year and keep the track clear of leaves. Replacement wheel sets are cheap and stocked.
Power point	About 100 above the deck on the studio wall; the leads now leave through 60 mm holes 160 above the deck. Measure its position along the wall. An outdoor outlet must be weatherproof (IP-rated).
Rear door set-back	With bypass doors one door has to run behind the other: the laundry door sits about 26 behind the flush fridge door.
Door flatness	1916 tall formply doors rely on their balanced film faces to stay flat. Keep both faces sealed; if one bows, a black aluminium angle screwed up its back edge fixes it.
Washer door	With the laundry door open, the parked door sits just left of the washer and 60 to 80 back from its face. A left-hinged washer door opened past about 120 degrees will touch it. Prefer a right-hinged washer, or open it to 90 degrees.

ASA print	Qty	Size	Notes
04_cable_grommet.stl	2	82 × 82 × 20	Split grommet for the two 60 mm lead holes (C2, C3). Smooth bore protects the leads. Print flange down. HOLD: confirm the plugs fit through 48 mm or enlarge the hole first.
05_edge_pad.stl	13	80 × 24 × 4	4 mm pad under the ply edges (13 of them). The box reads as flush to the deck but the ply never sits in water and the deck boards can dry. Bed in sealant, 1 x 10g x 25 into the deck. Print 100% infill.
06_rear_louvre.stl	12	230 × 120 × 30	Flush louvre insert for each back opening (12). Only the 3 mm flange stands proud of the back. Blades slope 45 degrees down and out and overlap, so there is no straight path in for rain; water on a blade runs down and out. Push into the 210 x 100 opening from outside, bedded in sealant, 2 x 6g x 16. Staple stainless mesh across the inside. Print on its flange, no supports (45 degree blades).
07_door_top_shoe.stl	4	80 × 21 × 41	Door top shoe, 2 per door (4). Glue (PU) and screw onto the door top edge, leg down the back face, 2 x 6g x 16 through the leg. The Quick-Fix wheel bracket then screws down through the 6 mm seat into the 17 mm door edge (use screws 6 mm longer than the bracket's own). Gives a 21 mm wide seat for a bracket rated for 19 mm+ doors. Print seat-down, 100% infill.